Homework 10
Introduction to Complex Analysis
(MATH 4230-40), Spring 2022

Section 4.1 - Calculation of Residues

1. Find the residues of the following functions at the indicated points:

   (a) $\dfrac{e^z - 1}{\sin z}$, $z_0 = 0$

   (b) $\dfrac{1}{e^z - 1}$, $z_0 = 0$

   (c) $\dfrac{z + 2}{z^2 - 2z}$, $z_0 = 0$

   (d) $\dfrac{1 + e^z}{z^4}$, $z_0 = 0$

   (e) $\dfrac{e^z}{(z^2 - 1)^2}$, $z_0 = 1$

2. Explain what is wrong with the following reasoning:
   Let $f(z) = \dfrac{1 + e^z}{z^2} + \dfrac{1}{z}$. *Since* $f(z)$ *has a pole at* $z = 0$, *the residue at that point is the coefficient of* $\dfrac{1}{z}$, *namely* 1.
   Then, compute the residue correctly.

3. If f_1 and f_2 have residues r_1 and r_2 at z_0, show that the residue of $f_1 + f_2$ at z_0 is $r_1 + r_2$.